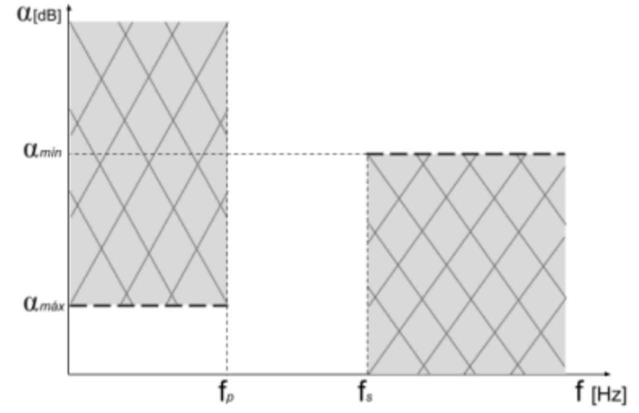


αmáx [dB]	ωp [r/s]	ωs [r/s]
1	1	2,5

1. Obtener la función transferencia T(s) de Bessel para N: 2, 3 y 4 normalizados para $D(\omega = 0) = 1s$ utilizando el método de Storch: ver *Schaumann, R. - Van Valkenburg, Mac E., Design of Analog Filters, Capítulo 10: Delay Filters. Sección 10.2: Bessel-Thomson Response. Página 403.*

N=2

$$\operatorname{cotgh} \approx \frac{1}{s} + \frac{1}{\frac{3}{s}} = \frac{3+s^2}{3s} \longrightarrow B_2 = s^2+3s+3$$

$$T_2(s) = \frac{B_2(0)}{B_2(s)} = \frac{3}{s^2+3s+3}$$

N=3

$$\operatorname{cotgh} \approx \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s}}} = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{s}{5}} = \frac{1}{s} + \frac{5s}{15+s^2}$$

$$= \frac{5s^2+15+s^2}{s^3+15s}$$

$$B_3 = s^3+6s^2+15s+15$$

$$T_3(s) = \frac{15}{s^3+6s^2+15s+15}$$

N=4

$$B_4 = (2\cdot 4-1)\cdot B_3 + s^2 B_2 = 7\cdot(s^3+6s^2+15s+15) + s^2\cdot(s^2+3s+3)$$

$$= 7s^3+42s^2+105s+105+s^4+3s^3+3s^2 = s^4+10s^3+45s^2+105s+105$$

$$T_{4(s)} = \frac{105}{s^4 + 10s^3 + 45s^2 + 105s + 105}$$

2. Elegir la T(s) con el <u>mínimo orden</u> que cumpla con $\alpha_{máx} = 1dB$.

$N = 2$

$$T_{2(\omega)} = \frac{3}{-\omega^2 + 3 \cdot j\omega + 3} = \frac{3}{3 - \omega^2 + j\,3\omega}$$

$$|T_{2(\omega)}| = \frac{3}{\sqrt{(3-\omega^2)^2 + (3\omega)^2}} = \frac{3}{\sqrt{9 - 6\omega^2 + \omega^4 + 9\omega^2}} = \frac{3}{\sqrt{\omega^4 + 3\omega^2 + 9}}$$

$$|\alpha_{2(\omega_p=1)}| = \left(\frac{3}{\sqrt{1 + 3 + 9}}\right)^{-1} = \frac{\sqrt{13}}{3} \simeq 1{,}2$$

$$\alpha_{2(\omega_p=1)}\Big|_{dB} = 20 \log\left(\frac{\sqrt{13}}{3}\right) \simeq 1{,}597 \text{ dB}$$ NO CUMPLE

$N=3$

$$T_{3(\omega)} = \frac{15}{-j\omega^3 - 6\omega^2 + 15 \cdot j\omega + 15} = \frac{15}{-6\omega^2 + 15 + j(-\omega^3 + 15\omega)}$$

$$|T_{3(\omega)}| = \frac{15}{\sqrt{(15 - 6\omega^2)^2 + (15\omega - \omega^3)^2}}$$

$$|T_{3(\omega_p=1)}| = \frac{15}{\sqrt{(15-6)^2 + (15-1)^2}} = \frac{15}{\sqrt{81 + 196}} \simeq 0{,}901$$

$$\alpha_{(\omega_p=1)\,dB} = -20 \log\left(\frac{15}{\sqrt{277}}\right) = \boxed{0{,}903 \text{ dB}}$$ CUMPLE

$N = 4$

$$T_4(\omega) = \frac{105}{\omega^4 - j\omega^3 10 - 45\omega^2 + j\omega 105 + 105}$$

$$|T_4(\omega)| = \frac{105}{\sqrt{(\omega^4 - 45\omega^2 + 105)^2 + (-\omega^3 10 + \omega \cdot 105)^2}}$$

$$|T_4(\omega_p = 1)| = \frac{105}{\sqrt{(1 - 45 + 105)^2 + (-10 + 105)^2}} = \frac{105}{\sqrt{12.746}} \simeq 0{,}93$$

$$\alpha_4(\omega_p = 1)\Big|_{dB} = 20 \cdot \log\left(\frac{\sqrt{12476}}{105}\right) \simeq \boxed{0{,}54 \ dB}$$ cumple

con $n = 3$ ya cumple...

3. Evaluar el retardo de grupo $D(\omega = 2.5)$ y expresar en forma *porcentual [%]* el error o desviamiento respecto a $D(\omega = 0)$.

$$D(\omega) = -\frac{d \measuredangle T(\omega)}{d\omega}$$

$$\measuredangle T_3(\omega) = -tg^{-1}\left(\frac{-\omega^3 + 15\omega}{-6\omega^2 + 15}\right)$$

$$D(\omega) = -\frac{1}{1 + \left(\frac{15\omega - \omega^3}{15 - 6\omega^2}\right)^2} \cdot \left[\frac{(-3\omega^2 + 15) \cdot (15 - 6\omega^2) + 12\omega \cdot (-\omega^3 + 15\omega)}{(15 - 6\omega^2)^2}\right]$$

$$D(\omega) = \frac{(15 - 6\omega^2)^2}{(15 - 6\omega^2)^2 + (15\omega - \omega^3)^2} \cdot \frac{18\omega^4 - 45\omega^2 + 225 - 90\omega^2 - 12\omega^4 + 180\omega^2}{(15 - 6\omega^2)^2}$$

$$D(\omega) = \frac{\cancel{(15-6\omega^2)^2}}{(15-6\omega^2)^2+(15\omega-\omega^3)^2} \cdot \frac{6\cdot\omega^4 + 45\omega^2 + 225}{\cancel{(15-6\omega^2)^2}}$$

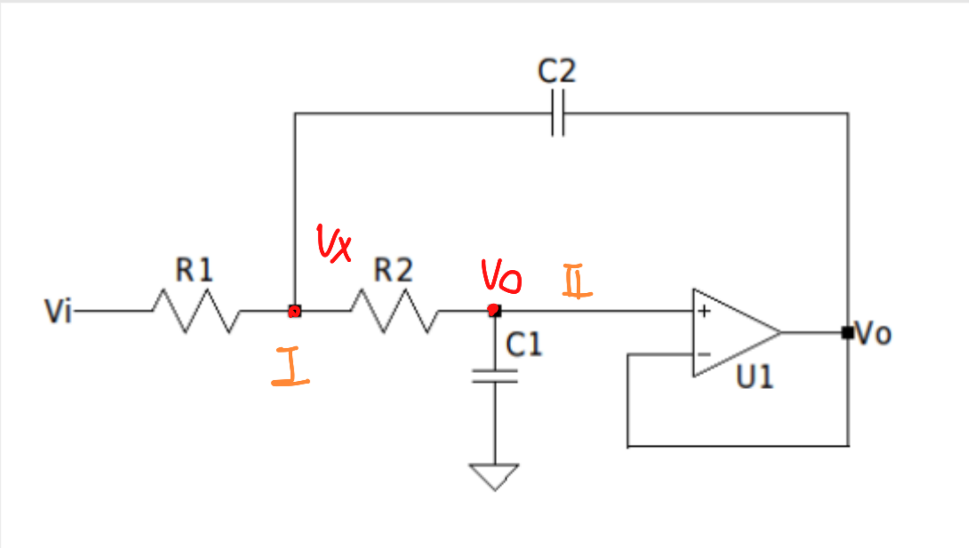
$$D_{(\omega=0)} = \frac{1}{225} \cdot \frac{225}{1} = \boxed{1}$$

$$D_{(\omega_s=2{,}5)} = \frac{6\cdot(2{,}5)^4 + 45(2{,}5)^2 + 225}{(15-6\cdot 2{,}5^2)^2+(15\cdot 2{,}5-2{,}5^3)^2} = \simeq \boxed{0{,}752}$$

finalmente...

$$\varepsilon_{\%} = \frac{D_{(0)} - D_{(\omega_s)}}{D_{(0)}} \cdot 100\% \cong 25\%$$

4. Sintetizar el circuito **NORMALIZADO** con estructuras *Sallen-Key con K=1 (real, negativa y unitaria).*



como necesito un orden 3, podría usar un sallen-key junto a un R-C, o implementar directamente el orden 4. Por cómo lo plantee en el ej. anterior, utilizaré la primera alternativa.

Procedo a obtener $T_{(s)}$ del Sallen-key

I: $V_x \cdot (G_1 + G_2 + SC_2) - V_o \cdot (SC_2 + G_2) - V_i \cdot G_1 = 0$

II: $V_o \cdot (G_2 + SC_1) - V_x \cdot G_2 = 0$

$$V_x = V_o \cdot \left(1 + \frac{SC_1}{G_2}\right)$$

I: $V_o\left(1+\frac{SC_1}{G_2}\right)\cdot(G_1+G_2+SC_2) - V_o(SC_2+G_2) - V_i\cdot G_2 = 0$

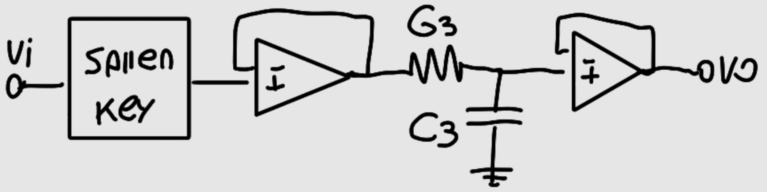
$$V_o\cdot\left(G_1+G_2+SC_2+SC_1\frac{G_1}{G_2}+S\cdot C_1 + S^2\cdot C_1\cdot\frac{C_2}{G_2} - SC_2 - G_2\right) = V_i\cdot G_2$$

$$V_o\cdot\left(G_1 + SC_1 + SC_1\frac{G_1}{G_2} + S^2 C_1\frac{C_2}{G_2}\right) = V_i\cdot G_1$$

$$\frac{V_o}{V_i} = T_{(s)} = \frac{G_1}{\frac{G_1\cdot G_2 + SC_1G_2 + SC_1\cdot G_1 + S^2C_1C_2}{G_2}}$$

$$T_{(s)} = \frac{G_1\cdot G_2 / C_1\cdot C_2}{S^2 + S\,\frac{G_2+G_1}{C_2} + \frac{G_1\cdot G_2}{C_1\cdot C_2}}$$

y si le coloco un filtro RC a la salida...



$$T_{T(s)} = \underbrace{\frac{\frac{G_1G_2}{C_1C_2}}{S^2+S\,\frac{G_1+G_2}{C_2}+\frac{G_1G_2}{C_1C_2}}}_{T_1} \cdot \underbrace{\frac{\frac{G_3}{C_3}}{S+\frac{G_3}{C_3}}}_{T_2}$$

Por otro lado, descomponiendo la $T_{3\,Bessel}(s)$ en 2 polinomios...

$$T_{3(s)} = \frac{15}{s^3+6s^2+15s+15} = \frac{A}{s+A} \cdot \frac{C}{s^2+Bs+C}$$

$$\frac{15}{s^3+6s^2+15s+15} = \frac{AC}{s^3+s^2(A+B)+s(C+BA)+CA}$$

$$CA = 15 \longrightarrow C = 15/A$$

$$C+BA = 15 \longrightarrow \frac{15}{A} + (6-A)\cdot A = 15$$

$$6 = A+B \longrightarrow B = 6-A$$

$$15 + 6A^2 - A^3 - 15A = 0$$

$$A = 2{,}32$$

$$B = 3{,}68$$

$$C = 6{,}46$$

Entonces...

$$A = \frac{G_3}{C_3}, \quad B = \frac{G_1+G_2}{C_2}, \quad C = \frac{G_1 G_2}{C_1 C_2}$$

$$C_3 = C_2 = 1, \quad G_1 = G_2$$

$$G_3 = 2{,}32, \quad G_1 = G_2 = 1{,}84, \quad C_1 = 0{,}524$$

Finalmente

$$T_{(s)} = \frac{\frac{G_1^2}{C_1}}{S^2 + 2G_1 + \frac{G_1^2}{C_1}} \cdot \frac{G_3}{S + G_3}$$

$C_2 = C_3 = 1$, $G_1 = G_2 = 1{,}84$, $G_3 = 2{,}32$, $C_1 = 0{,}524$

- +10 💎 DESNORMALIZAR los componentes para obtener un $D(\omega = 0) = 200\mu s$.

Para $D(\omega=0)$ de $200\mu s$...

$\Omega\omega = \Omega\omega$

$$T_{(\omega)} = \frac{15 \cdot \Omega\omega^3}{-6\omega^2 \cdot \Omega\omega + 15\Omega\omega^3 + j(-\omega^3 + 15\omega \cdot \Omega\omega^2)}$$

$$\measuredangle T(\omega) = -tg^{-1}\left(\frac{-\omega^3 + 15\omega \cdot \Omega\omega^2}{-6\omega^2\Omega\omega + 15\Omega\omega^3}\right)$$

$$D(\omega) = \frac{1}{1 + \left(\frac{15\omega\Omega\omega^2 - \omega^3}{15\Omega\omega^3 - 6\omega^2\Omega\omega}\right)^2} \cdot$$

$$\frac{(-3\omega^2 + 15\Omega\omega^2) \cdot (15\Omega\omega^3 - 6\omega^2\Omega\omega) + 12\omega\Omega\omega(-\omega^3 + 15\omega\Omega\omega^2)}{(15\Omega\omega^3 - 6\omega^2\Omega\omega)^2}$$

$$D_{(0)} = \frac{1}{1+0^2} \cdot \frac{225\,\Omega w^5}{(15\,\Omega w^3)^2} = 200\mu s$$

$$200\mu s = \Omega w^{-1}$$

$$\Omega w = 5000$$

Por ende...

$$T_{(s)} = \frac{\frac{G1}{C1} \cdot \Omega w}{S^2 + \Omega w \cdot 2 \cdot G1 + \frac{G1^2}{C1} \cdot \Omega w} \cdot \frac{G3 \cdot \Omega w}{S + G3 \cdot \Omega w}$$

$C_2 = C_3 = 200\mu F$, $G1 = G2 = 1{,}84$, $G3 = 2{,}32$, $C_1 = 104{,}8\mu f$

$$T_{(s)} = \frac{87{,}79 \cdot 10^6}{S^2 + 18.400S + 87{,}79 \cdot 10^6} \cdot \frac{11.600}{S + 11600}$$